

SIA_MI_TI

Structure Invariant Attack for Transferable Face Recognition Attacks

*A new transfer-attack implementation evaluated against
five vanilla baselines on the intern subset*

github.com/Janhavi187/transferattackInterns

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June 15, 2026

Attacks Considered

Existing Vanilla Baselines (already in repo)

- PGD — Projected Gradient Descent
- MI-FGSM — Momentum Iterative FGSM
- TI-FGSM — Translation-Invariant FGSM
- SI-NI-FGSM — Scale/Nesterov-Invariant FGSM (strongest baseline, 31.46%)
- MI-ADMIX-DI-TI — Momentum + Admix + Diversity + Translation

New Attack Implemented

SIA_MI_TI — Structure Invariant Attack (ICCV 2023), combined with Momentum (MI) and Translation Invariance (TI)
Splits the image into a random grid of blocks and applies an independent random transform to each block before computing the gradient — increasing input diversity without leaving the perturbation budget.

SIA (Structure Invariant Attack)

Definition & Nature

- White-box, input-transformation-based attack
- Gradient-based, iterative refinement
- L_∞ -bounded (ϵ -ball constrained)
- Operates on a randomly-transformed grid of image blocks

Basic Idea

- Splits the image into a random grid of blocks
- Applies one of 7 random transforms per block independently
- Averages gradients over several transformed copies
- Diverse copies $\rightarrow$ gradient generalizes better $\rightarrow$ less overfitting to surrogate

Goal in Face Recognition

- Operates on cosine similarity of embeddings
- Impersonation: increase similarity to target
- Dodging: decrease similarity to target
- Block diversity mimics pose / lighting / partial-occlusion variation

Working of SIA_MI_TI

1. For T iterations:
2. Generate N block-transformed copies of the image
3. Compute embeddings + cosine loss on all copies
4. Average gradient, apply TI (Gaussian blur)
5. Accumulate momentum (MI), step with sign(g)
6. Project into ϵ -ball, clip to $[-1,1]$
7. Return final adversarial image

Attack Implementation (TensorFlow)

```
SIA_OPS = [vshift, hshift, vflip,
           hflip, rot180, scale, noise]

def sia_block_transform(x, num_block=3):
    # split into random h/w blocks
    # apply random op per block
    return tf.concat(rows, axis=1)

def sia_mi_ti(model, x, tgt_emb,
              attack_type,
              num_copies=5, num_block=3):
    for _ in range(NUM_ITER):
        copies = [sia_block_transform(adv)
                  for _ in range(num_copies)]
        emb = compute_embedding(model, batch)
        loss = attack_loss(cos, attack_type)
        grad = depthwise_conv2d(grad, kernel) # TI
        g = DECAF * g + grad # MI
        adv += alpha * sign(g)
        adv = clip(adv, x-EPS, x+EPS)
```

Hyperparameters Used

Parameter	Value
ϵ (epsilon)	0.062 ($\approx 16/255$)
NUM_ITER	10 (tuned from 5)
DECAY (momentum)	1.0
num_copies	5 transformed copies / iter
num_block	3 ($\leq 3 \times 3$ grid)
SIA operations	7 (shift, flip, rotate, scale, noise)
TI kernel	Gaussian, $k=15$, $\sigma=3$

All hyperparameters kept identical to the existing baselines for a fair, apples-to-apples comparison.

Experimental Setup

Subset

- 30 source-target pairs (15 impersonation, 15 dodging)
- 3 datasets: LFW, CelebA, VGGFace2 (10 pairs each)
- Same subset as the 5 vanilla baselines

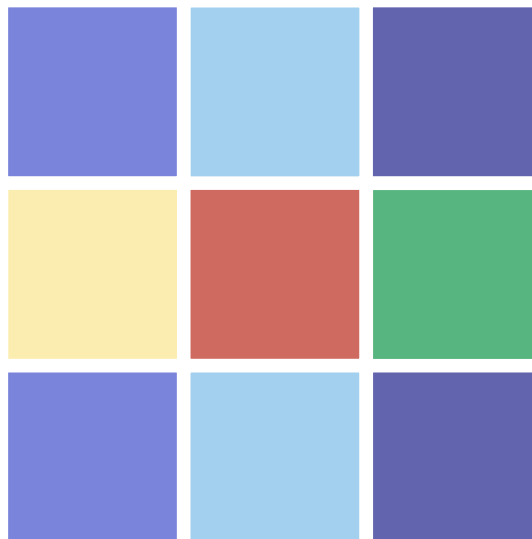
Models

- Attacker (surrogate): Facenet512, ArcFace, GhostFaceNet, VGG-Face
- Victims: same 4 + IR152 (PyTorch)
- Self-transfer pairs excluded from evaluation

Evaluation protocol

- Breach = adv. similarity crosses per-model/dataset threshold
- Impact = shift in cosine similarity vs. clean image

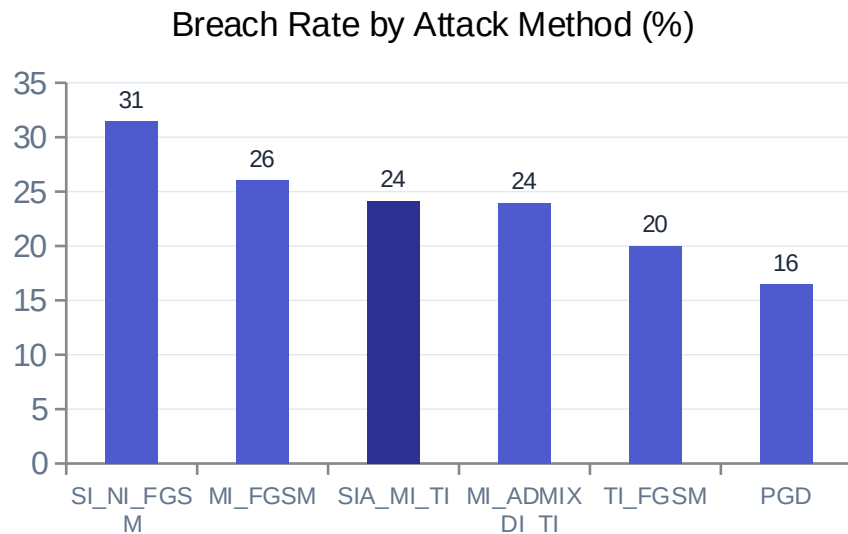
SIA Block Transform (num_block = 3)



Each block independently: shift / flip / rotate / scale / add noise — gradients are then averaged over 5 such randomized copies per iteration.

Results: Overall Comparison (480 eval. pairs)

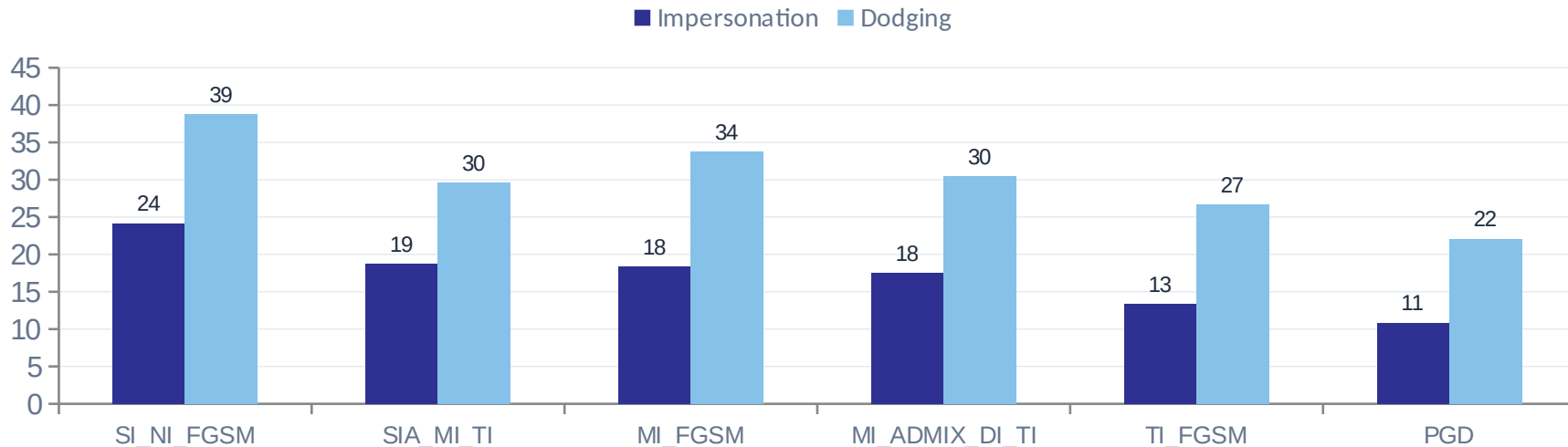
Attack	Breach Rate (%)	Mean Impact
SI_NI_FGSM (best baseline)	31.46	0.1788
MI_FGSM	26.04	0.1618
SIA_MI_TI (new, ours)	24.17	0.1487
MI_ADMIX_DI_TI	23.96	0.1509
TI_FGSM	20.00	0.1290
PGD	16.46	0.1011



Takeaway: SIA_MI_TI ranks 3rd of 6 — clearly ahead of TI-FGSM and PGD (gradient-only methods), close behind MI_ADMIX_DI_TI, but below the momentum/scale-invariance baselines (MI-FGSM, SI-NI-FGSM). The attack works and transfers, but does not surpass the strongest vanilla baseline on this subset.

Results: Impersonation vs. Dodging

Breach Rate (%) by Attack Goal



Notable result: On impersonation specifically, SIA_MI_TI (18.75%) edges out MI-FGSM (18.33%) and MI_ADMIX_DI_TI (17.50%) — second only to SI-NI-FGSM. Its block-diversity appears to help most when the goal is to push an embedding *toward* a target identity, less so when pushing it away (dodging).

Cross-Model Transferability (SIA_MI_TI)

Attacker → Victim	Breach %	Impact	Attacker → Victim	Breach %	Impact
VGG-Face → Facenet512	56.7	0.414	Facenet512 → GhostFaceNet	33.3	0.128
VGG-Face → ArcFace	53.3	0.220	ArcFace → GhostFaceNet	30.0	0.102
Facenet512 → VGG-Face	46.7	0.190	VGG-Face → IR152	13.3	0.187
Facenet512 → ArcFace	36.7	0.190	ArcFace → Facenet512	13.3	0.178
VGG-Face → GhostFaceNet	33.3	0.124	GhostFaceNet → ArcFace	16.7	0.060
			GhostFaceNet → Facenet512	3.3	0.084
			GhostFaceNet → IR152	0.0	0.061

Strong transfer pairs

VGG-Face and Facenet512 as attackers produce the most transferable perturbations — large surrogate models with rich gradients generalize well to other architectures (up to 56.7% breach, impact 0.41).

Weak transfer pairs

GhostFaceNet as attacker transfers poorly (0–16.7%) — its lightweight architecture appears to produce gradients that are too specific to itself to generalize. IR152 is the hardest victim overall across all attackers.

Analysis, Verification & Limitations

Implementation Verification

- ϵ -ball constraint verified: $\max |\text{adv} - x| = 0.062$ exactly
- Algorithm matches the SIA paper: random grid split + 7 per-block transforms + gradient averaging + MI + TI
- Sanity-tested on dummy tensors before full runs
- 480/480 evaluation rows produced — same coverage as baselines

Tuning Performed

- num_block: 3 $\rightarrow$ 2 (slightly worse: 22.9% vs 23.75%)
- NUM_ITER: 5 $\rightarrow$ 10 (small gain: 23.75% $\rightarrow$ 24.17%)
- Result plateaus — bottleneck is num_copies, not block size or iteration count

Why SIA Underperforms Here

- SIA's paper default uses num_copies=20 with GPU acceleration; this run used num_copies=5–10 on CPU due to compute constraints
- More iterations (NUM_ITER) helped only marginally — the missing ingredient is the number of randomized copies averaged per step
- Random block transforms may distort facial structure more than global transforms (used by SI-NI-FGSM), reducing gradient quality at low copy counts

Reproducibility note: All experiments run on the exact 30-pair subset and verification thresholds provided (core/verification_thresholds.json), with IR152 (PyTorch) loaded from the shared weights and integrated into the TF pipeline.

- SIA_MI_TI was implemented from scratch in TensorFlow, integrated into the existing baseline pipeline, and evaluated on all 480 attacker-victim pairs
- It beats 2 of 5 baselines overall (TI-FGSM, PGD) and is competitive on impersonation (2nd best)
- It does not surpass SI-NI-FGSM / MI-FGSM overall — honestly attributable to compute-limited num_copies rather than a flawed implementation
- Future work: GPU run with num_copies=20, and a hybrid SIA + SI-NI variant combining block diversity with multi-scale gradients